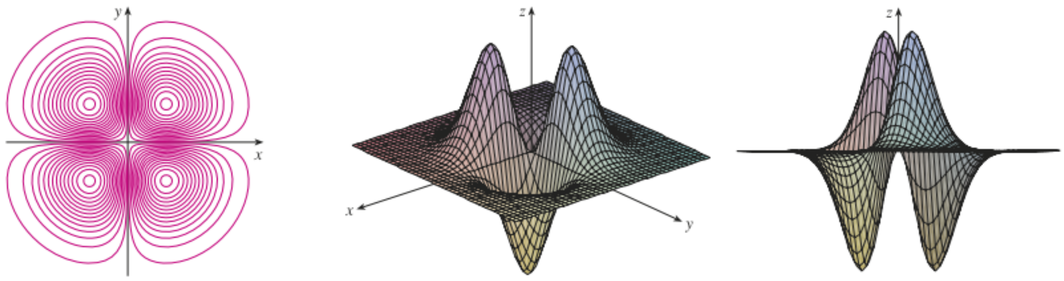
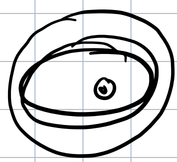
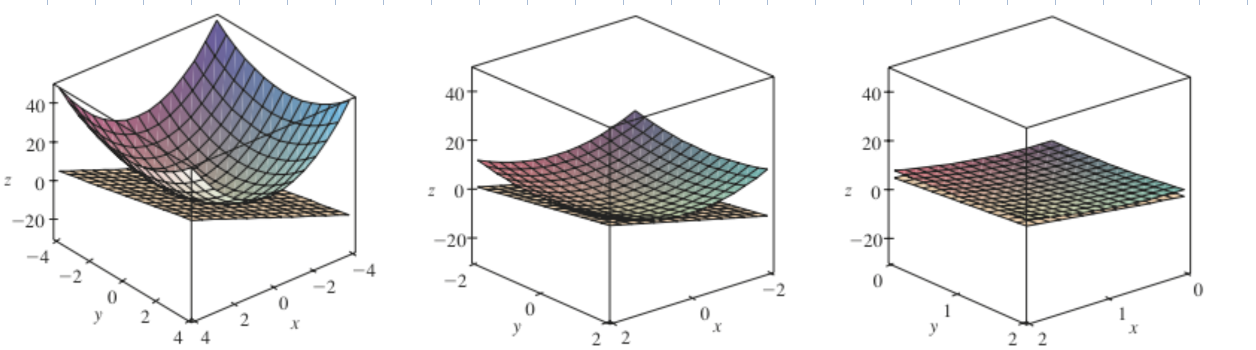


(a) Level curves of $f(x, y) = -xye^{-x^2-y^2}$

(b) Two views of $f(x, y) = -xye^{-x^2-y^2}$



$f(x)\ xe^{-x^2}$



(a)

(b)

(c)

$z = (2?)\, x^2 + y^2$

$\boxed{\begin{array}{l} x = 1 \\ y = 1 \end{array}}$

Textbooks

1) Online text (from classroom)
2) Calculus: Early Transcendentals (J. Stewart).
3) Scribble notes.

$$f: \mathbb{R}^n \to \mathbb{R}^m$$

- Functions with more than one variable ($m=1$, $n>1$)
- Functions with more than one value. (vector-valued) ($m>1$, $n=1$)

① How to describe a plane in $\mathbb{R}^3$:

Null space

$$ax + by + cz = d \quad \Big\} \quad \begin{pmatrix} a & b & c \end{pmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = d$$

for some $a, b, c, d \in \mathbb{R}$.

② How to describe a line in $\mathbb{R}^3$.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + t \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

different values of t sweep the line.

image space.

$$\begin{cases} a_1 x + b_1 y + c_1 z = 0 \\ a_2 x + b_2 y + c_2 z = 0 \end{cases}$$

$$\underbrace{\begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{pmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{} = 0$$

③ $x^2 + y^2 = 1$ cylinder

(7) $x^2+y^2 = z$ $\quad z \geq 0$ $\quad f(x,y) = x^2+y^2$

Visulizing 2-variable fns

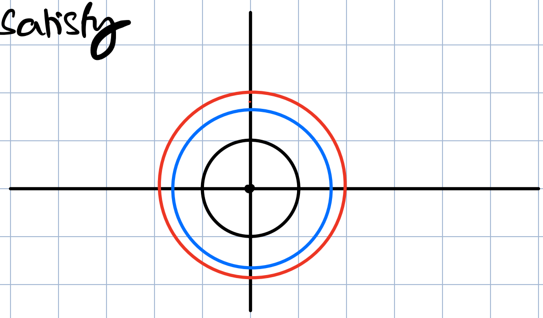
$\rightarrow$ plot in 3d.

Level curves:

plot $f(x,y) = K$.

all x,y that satisfy

$x^2+y^2=K$



$K=1$

$K=2$

$K=3$

Excercise Draw / visualise $f(x,y) = x^2$

Background

1) Calculus of single variable fns

2) Dot product.

3) Cross product. $(\mathbb{R}^3)$

4) min / max of single variable functions.

$a = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$ $\quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

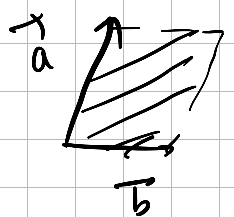
$\vec{a}$

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$\vec{a}\times\vec{b} = -\vec{b}\times\vec{a}$

$\vec{a}\times(\vec{b}+\vec{c})$

$= \vec{a}\times\vec{b} + \vec{a}\times\vec{c}$



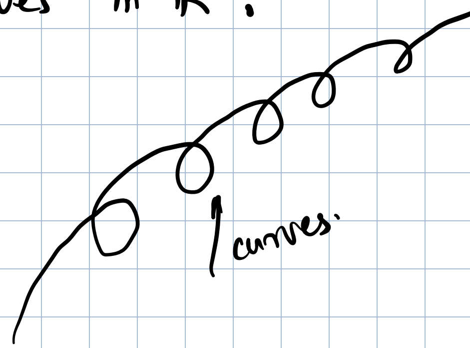
Area of parallelogram formed by $\vec{a}$ & $\vec{b}$

is $|\vec{a}\times\vec{b}|$

Vector valued functions:

$f: \mathbb{R} \to \mathbb{R}^n$: curves in $\mathbb{R}^n$.

$\underbrace{\vec{r}(t)}$:

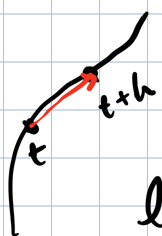


Let $\underbrace{\vec{r}(t)} = \begin{bmatrix} r_1(t) \\ r_2(t) \\ \vdots \\ r_n(t) \end{bmatrix}$

then $\vec{r}(t)$ is continuous / differentiable (respectively) if & only if

$r_1, r_2, \ldots, r_n$ are continuous.

$\vec{r}(t) = \cos t\, \vec{i} + \sin t\, \vec{j} + t\, \vec{K}$ — (a) Helix.



$$\lim \frac{\vec{r}(t+h) - \vec{r}(t)}{h} = \vec{r}'(t)$$

$f(x)$

$$\frac{f(x+h) - f(x)}{h}$$

Obtained by taking component-wise derivatives

$a_n \to a$

(a) $\vec{r}'(t) = -\sin t\, \vec{i} + \cos t\, \vec{j} + \vec{K}$.

$\|a_n - a\| < \epsilon$

$\vec{r}(t)$: position

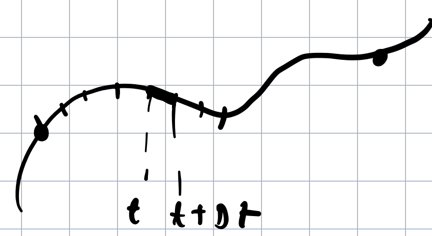
Properties:

$\vec{r}'(t)$: velocity $\quad |\vec{r}'(t)|$: speed

① $\vec{r}'(t)$ is the tangent to the curve at time t.

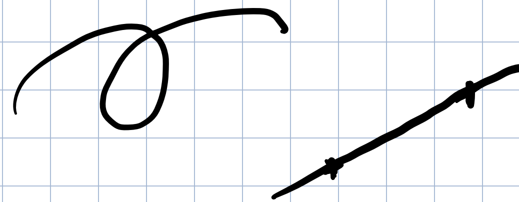
② Arc-length of arc from $t=a$ to $t=b$ is given by

$$\int_{t=a}^{t=b} |\vec{r}'(t)|\, dt$$



$$s(t) = \int^{t} |\vec{r}'(t)|\, dt$$

$$\frac{ds}{dt} = |r'(t)|$$



③ curvature :

$$T(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}$$

Tangent direction as function of time.

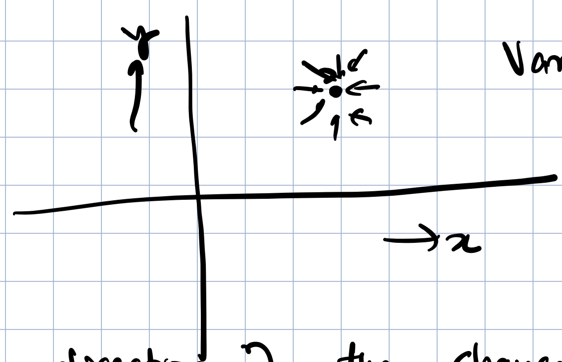
the curvature is defined as derivative of tangent.

Multivariable functions $f : \mathbb{R}^n \to \mathbb{R}$

$x_1 \quad x_2 \; - - \; x_n$

Continuity of two variable functions : $\min f(x_1, x_2, -, x_n)$

$f(x, y)$



Variable plane

$f(x)$

$f'(x) = 0$

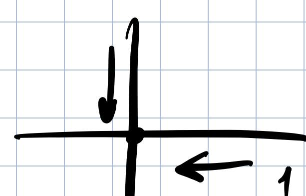
No matter the direction of the change, small changes to variable lead to small changes in function.

$\lim_{x \to 0^+}$

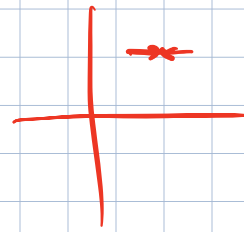
$\lim_{x \to 0^-}$

Ex: $f(x,y) = \dfrac{x^2 - y^2}{x^2 + y^2}$

Does $\lim_{(x,y)\to(0,0)} \dfrac{x^2 - y^2}{x^2 + y^2}$ exist ?



Partial & directional derivatives.



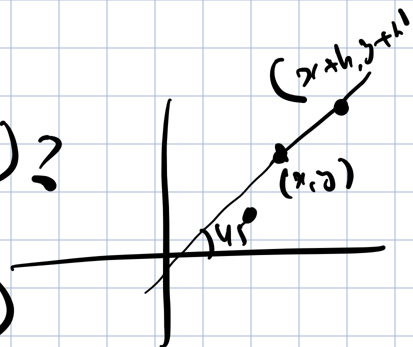
$$\frac{\partial f}{\partial x}(x,y) = \lim_{h\to 0} \frac{f(x+h,y) - f(x,y)}{h}$$

$f(x,y) = xy$

$\frac{\partial f}{\partial x}(x,y) = y.$

$$\frac{\partial f}{\partial y}(x,y) = \lim_{h\to 0} \frac{f(x,y+h) - f(x,y)}{h}$$

what is derivative along the direction $(x=y)$?



$$\lim_{h\to 0} \frac{f(x+h,y+h) - f(x,y)}{h}$$

$$\frac{f(x+h,y+h) - f(x+h,y) + f(x+h,y) - f(x,y)}{h}$$

$$\frac{\partial f}{\partial y}(x,y) + \frac{\partial f}{\partial x}(x,y)$$

$f: \mathbb{R}^n \to \mathbb{R}.$

$f: \mathbb{R}$

Let $\vec{a}$ be a unit vector in the variable plane.

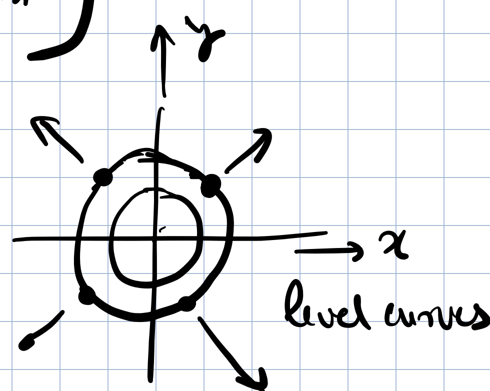
Suppose the partial derivatives ∂f are continuous, then the directional derivative along the direction $\vec{a}$ is defined as

$$\underbrace{a_1 \frac{\partial f}{\partial x_1} + a_2 \frac{\partial f}{\partial x_2} + \cdots + a_n \frac{\partial f}{\partial x_n}}_{\vec{a} \cdot \vec{\nabla f}}$$

$$\vec{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \qquad \vec{\nabla f} = \begin{bmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \\ \vdots \\ \partial f / \partial x_n \end{bmatrix}$$

$$f(x, y) = x^2 + y^2$$

$$\vec{\nabla f} = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$



Gradients are normal / perpendicular to level curves.

Tod 16: Directional derivative,

for a $f: \mathbb{R}^2 \to \mathbb{R}$, the directional derivative along direction $\vec{a}$,

$$\lim_{h \to 0} \frac{f(x + a_1 h, y + a_2 h) - f(x, y)}{h}$$

$$\vec{a} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

$$= \underbrace{a_1 \frac{\partial f}{\partial x_1} + a_2 \frac{\partial f}{\partial x_2}}$$

$$= \vec{a} \cdot \vec{\nabla f}$$

∂f

$$\nabla f(x,y) = \begin{bmatrix} \partial f/\partial x \\ \partial f/\partial y \end{bmatrix}$$

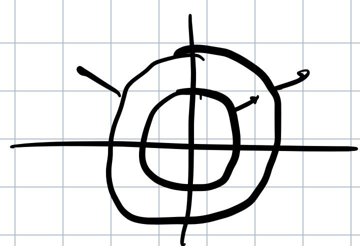
Gradient is the direction along which the rate of change is highest

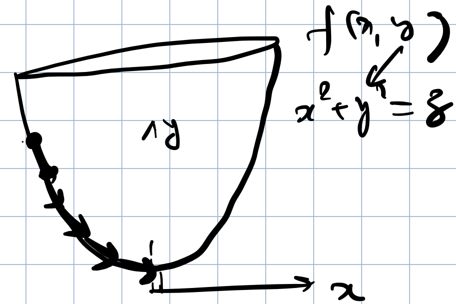
∇f : direction of steepest ascent

$-\nabla f$: direction of steepest descent.

Gradient is normal to level curves.

Gradient descent algo: Given a $f: \mathbb{R}^n \to \mathbb{R}$, find the minimum.





Start with arbitrary x_0

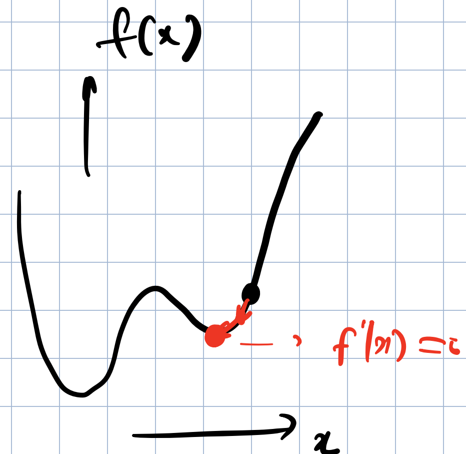
take a step in direction $-\nabla f$

$$x_1 = x_0 - \underbrace{\alpha}_{} \nabla f(x_0)$$

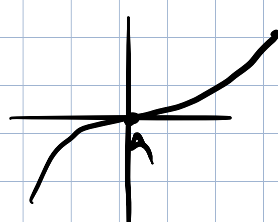
amount of movement

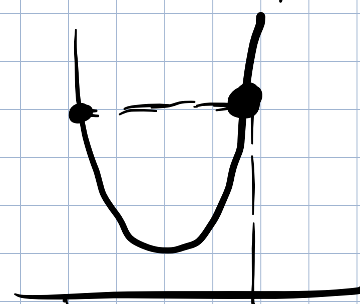
$$x_2 = x_1 - \alpha \nabla f(x_1)$$

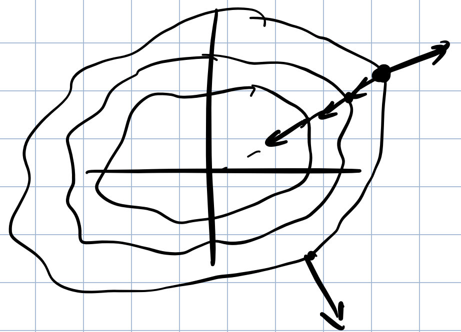
$\vdots$



- Can get stuck at local minima.
- performance dependence on initial point.







$$\boxed{f(a+h)-f(x) \approx h f'(x)}$$

First order approx:

$$f(x+a_1h,\ y+a_2h) \approx h a_1 \frac{\partial f}{\partial x} + h a_2 \frac{\partial f}{\partial y}$$

Tool 15(b):

$$f(x_0+\Delta x,\ y_0+\Delta y) \approx f(x_0,y_0) + \underbrace{\Delta x \frac{\partial f}{\partial x} + \Delta y \frac{\partial f}{\partial y}}_{\begin{pmatrix}\Delta x\\ \Delta y\end{pmatrix} \cdot \begin{bmatrix}\partial f/\partial x\\ \partial f/\partial y\end{bmatrix}}$$

when Δx & Δy are small.

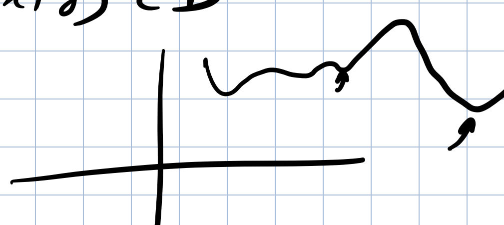
the equation of the tangent plane to the surface $z=f(x,y)$ at point (x_0,y_0) is given by

$$z = f(x_0,y_0) + (x-x_0)\frac{\partial f}{\partial x}\Big|_{x=x_0,\,y=y_0} + (y-y_0)\frac{\partial f}{\partial y}\Big|_{\substack{x=x_0\\ y=y_0}}$$

Local max/min:

We say that (x_0,y_0) is a local minima of f if in a small disc D that contains (x_0,y_0) in the interior,

$$f(x,y) \geq f(x_0,y_0) \quad \text{for all } (x,y) \in D.$$



(x_0, y_0) is a global minimum if

$f(x,y) \geq f(x_0, y_0)$ for all (x,y) in the domain of f.

One variable :

$f(x) = x^2$

$x \in [-2, 2]$

$x = 0$ LOCAL & GLOBAL MIN

$f(x) = x$

for $x \in [2,3]$

$x = 2$ GLOBAL MIN

NO LOCAL MIN.

Tool 16: (1) If (x_0, y_0) is a local minimum, then

$$\nabla f(x_0, y_0) = 0.$$

(2) If $\nabla f(x_0, y_0) = 0$ and the Hessian is positive definite, then (x_0, y_0) is a local minimum.

all eigenvalues positive.

Hessian matrix

2x2

$$\begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$$

$\frac{\partial f}{\partial x}$

$\frac{\partial f}{\partial y}$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

so long as they are continuous

$f(x,y) = x(10 - y^2)$

$$\nabla f = \begin{bmatrix} 10 - y^2 \\ -2xy \end{bmatrix}$$

$$H = \begin{bmatrix} 0 & -2y \end{bmatrix}$$

$$\begin{pmatrix} -2y & -2x \end{pmatrix}$$

$$f(x) = f(0) + x f'(0) + \underbrace{\frac{x^2}{2!} f''(0)} + \cdots$$

Tool 15 (c): Second order approx

$$f(x_0+\Delta x, y_0+\Delta y)$$

$$\approx f(x_0, y_0) + \Delta x \frac{\partial f}{\partial x} + \Delta y \frac{\partial f}{\partial y}$$

$$\underbrace{+ \frac{1}{2!} [\Delta x \quad \Delta y] \begin{bmatrix} \text{Hessian} \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}}$$

$$f(x,y) = x(10-y^2) \quad x_0 = 0 \quad y_0 = 1$$

$$f(x_0+\Delta x, y_0+\Delta y) \approx \underbrace{f(x_0, y_0)}_{=0} + [\Delta x \quad \Delta y] \begin{bmatrix} 9 \\ 0 \end{bmatrix}$$

$$+ \frac{1}{2} [\Delta x \quad \Delta y] \begin{bmatrix} 0 & -2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}$$

$$\Delta x \left(10 - (1+\Delta y)^2\right)$$

$$\approx \underbrace{9 \Delta x - 2 \Delta x \Delta y}$$

Find global min:

local min

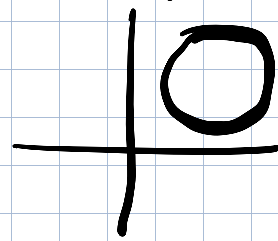
1) Compute gradient, set $\nabla f = 0$ to find critical points

2) For each of the critical points, compute the

Hessian and check its eigen values are +ve.

3) Check the values on the boundary.

4) Compare smallest value on boundary with value at local min.



$2x+3y$

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix}$$